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**SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
COMPUTER SCIENCE AND ENGINEERING PROGRAMME**

Day 3: List of Programs

Question 1: Apriori and FP-Growth Algorithm in WEKA

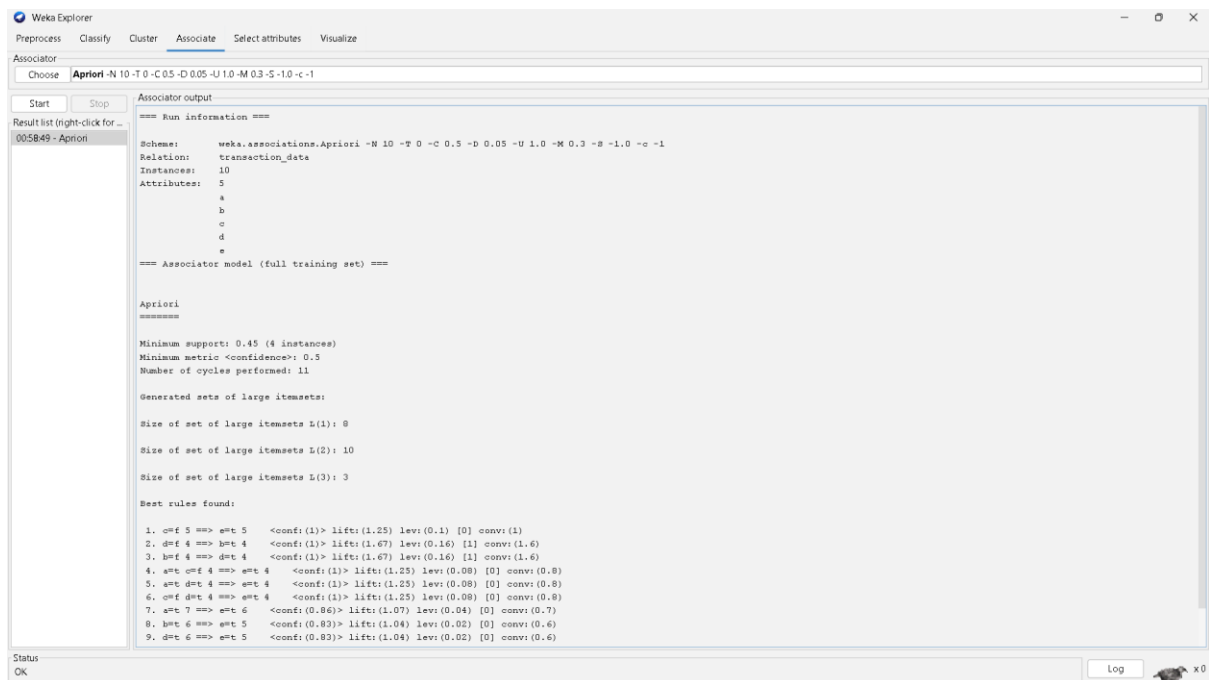
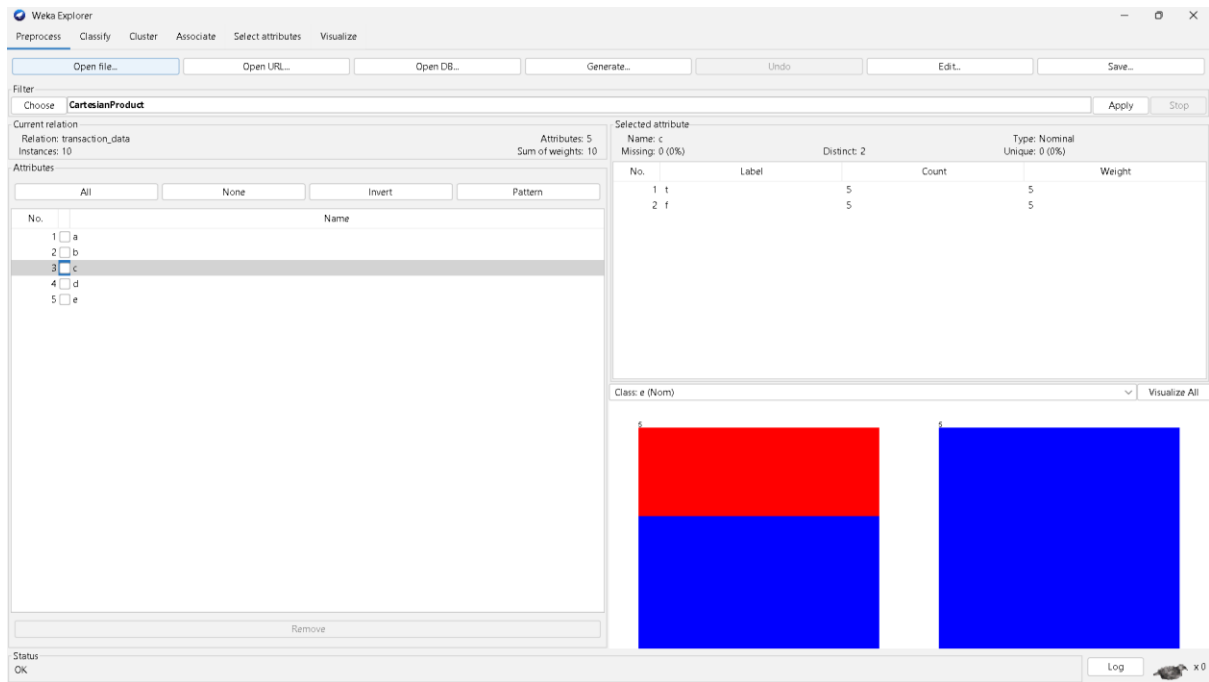
Software / Environment: WEKA Tool (Explorer → Associate)

Aim:

To implement the Apriori and FP-Growth association rule mining algorithms using the WEKA data mining tool with minimum support = 3 (30% / 0.3) and minimum confidence = 50% (0.5).

Procedure (Algorithm in WEKA):

1. Create an ARFF file (q1_dataset.arff) with binary nominal attributes ($\{t, ?\}$ or $\{t, f\}$) for items a, b, c, d, e based on the given transaction table.
2. Open **WEKA GUI Chooser** and click on **Explorer**.
3. Under the **Preprocess** tab, click **Open file...** and select q1_dataset.arff.
4. Navigate to the **Associate** tab.
5. **For Apriori:**
 - Click the text box next to the **Choose** button (which defaults to Apriori) to open the properties panel.
 - Set lowerBoundMinSupport = 0.3 (or 3 / 10).
 - Set minMetric (confidence) = 0.5.
 - Click **OK**, then click **Start**.
6. **For FP-Growth:**
 - Click **Choose** → select FPGrowth.
 - Set lowerBoundMinSupport = 0.3 and minMetric = 0.5.
 - Click **Start**.



Question 2: Market Basket Analysis

Aim:

To determine the maximum number of possible association rules and the maximum size of frequent itemsets from the given transaction dataset.

Algorithm:

1. Count the total number of unique items (d) present across all transactions.
2. Use the association rules combinatorial formula: $R = 3^d - 2^{d+1} + 1$ to find the total possible rules.
3. Identify the transaction with the maximum number of items to determine the upper bound on the size of any frequent itemset.

Calculation & Theory:

- Unique items list: {Milk, Beer, Diapers, Bread, Butter, Cookies} $\rightarrow d = 6$ unique items.
- (a) Maximum number of association rules:
$$R = 3^6 - 2^{6+1} + 1 = 729 - 128 + 1 = 602$$
- (b) Maximum size of frequent itemsets:

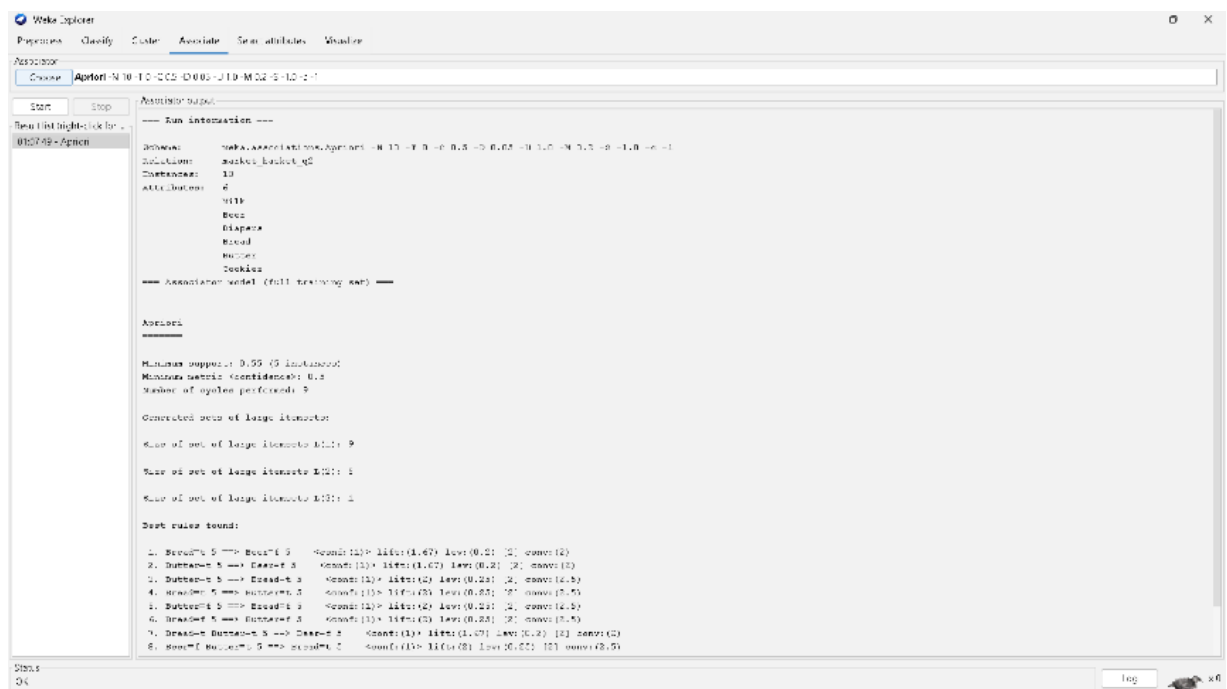
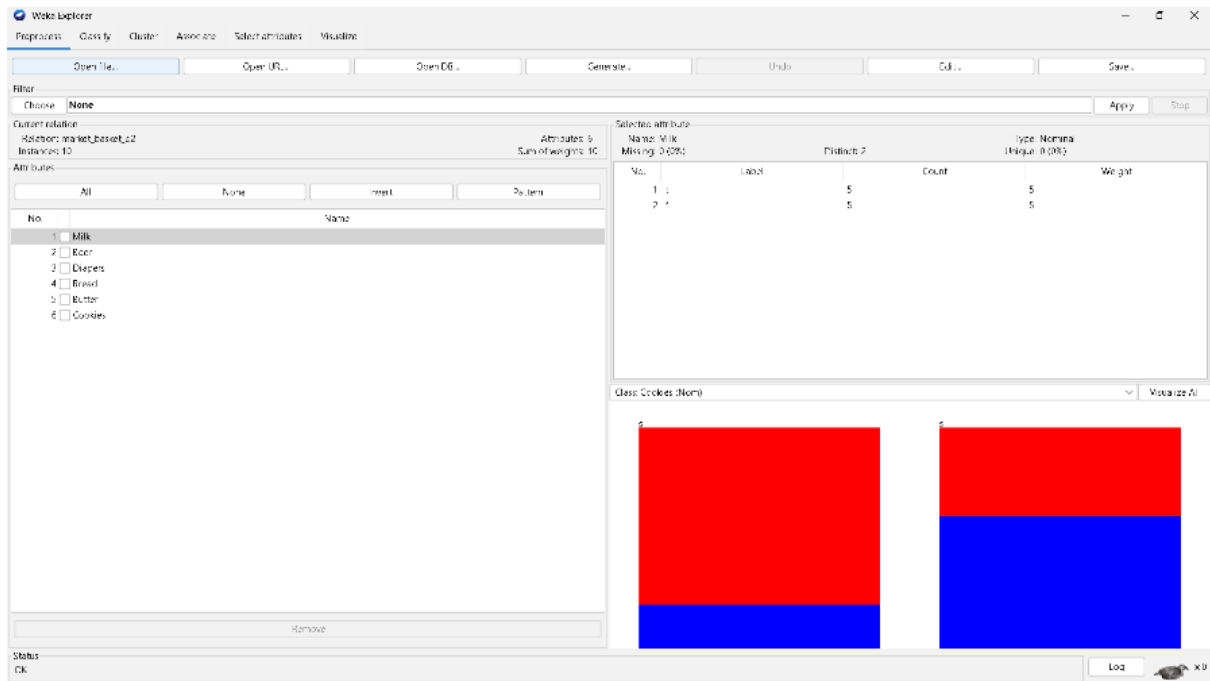
The longest transactions (Transactions 6 and 9) contain 4 items: {Milk, Diapers, Bread, Butter}.

Therefore, the maximum possible size of a frequent itemset is 4.

Result:

(a) Maximum number of association rules = 602

(b) Maximum size of frequent itemsets = 4



Question 3: Bayes Classification and Decision Tree in WEKA

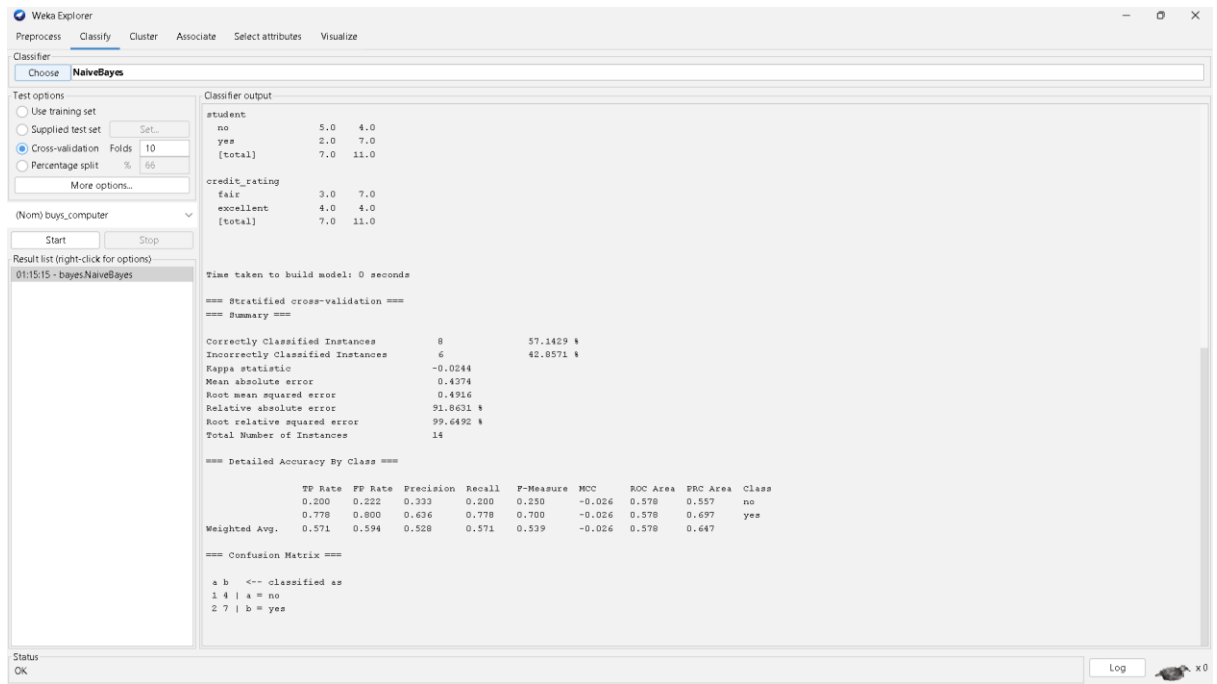
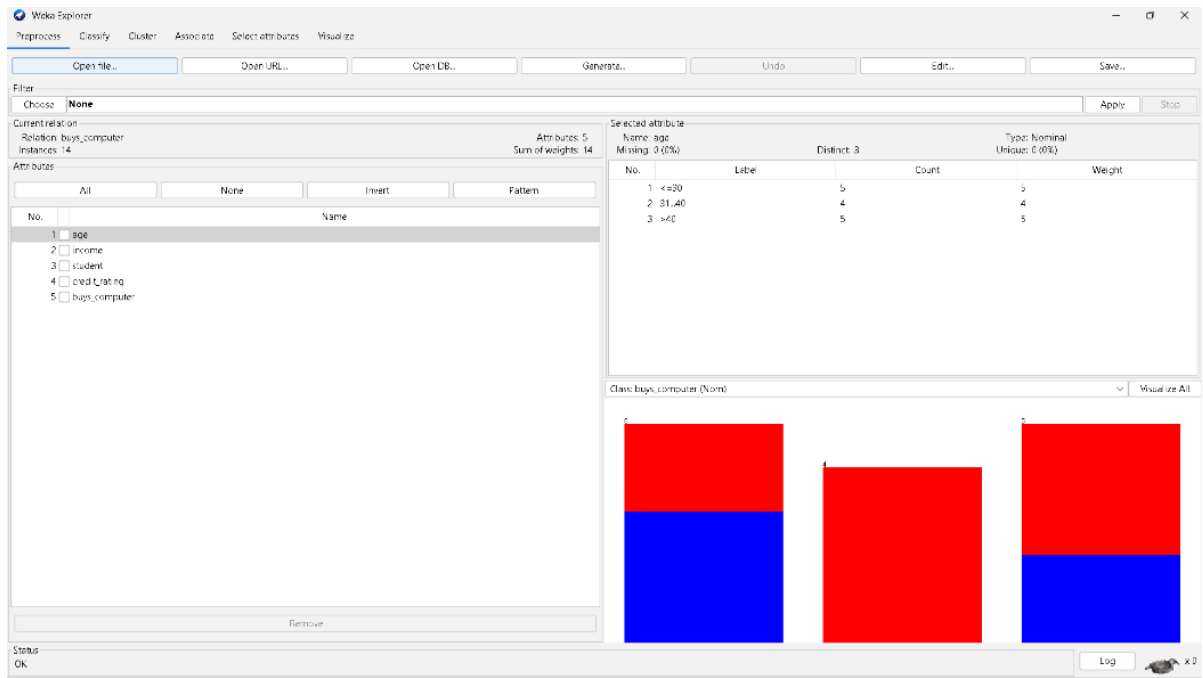
Software / Environment: WEKA Explorer (Preprocess & Classify tabs)

Aim:

To implement Naive Bayes and Decision Tree (J48) classification algorithms on the buys_computer dataset to predict whether a customer will buy a computer.

Procedure (Steps in WEKA):

1. Create a file named buys_computer.arff with nominal attributes age, income, student, credit_rating, and class attribute buys_computer.
2. Open **WEKA Explorer** and click **Open file...** under the **Preprocess** tab to load buys_computer.arff.
3. Click on the **Classify** tab at the top.
4. **For Naive Bayes:**
 - Click **Choose** → expand classifiers → expand bayes → select NaiveBayes.
 - Under *Test options*, select **Use training set**.
 - Click **Start** to view the accuracy and classification details.
5. **For Decision Tree (J48):**
 - Click **Choose** → expand classifiers → expand trees → select J48.
 - Under *Test options*, select **Use training set**.
 - Click **Start**.
 - Right-click the result entry in the *Result list* (bottom left) and click **Visualize tree** to view the tree structure.



Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **J48 - C 0.25 - M 2**

Test options:
☐ Use training set
☐ Supplied test set
☒ Cross-validation Folds: **10**
☐ Percentage split %: **66**
 More options...

(Nom) buys_computer
 Start Stop

Result list (right-click for options):
 01:15:15 - bayes.NaiveBayes
01:16:24 - trees.J48

Classifier output:

```

| student = no: no (3.0)
| student = yes: yes (2.0)
age = 31...40: yes (4.0)
age = >40
| credit_rating = fair: yes (3.0)
| credit_rating = excellent: no (2.0)

Number of Leaves : 5
Size of the tree : 8

Time taken to build model: 0 seconds

=== Stratified cross-validation ===
=== Summary ===
Correctly Classified Instances 7 50 %
Incorrectly Classified Instances 7 50 %
Kappa statistic -0.0426
Mean absolute error 0.4167
Root mean squared error 0.5984
Relative absolute error 87.5 %
Root relative squared error 121.2987 %
Total Number of Instances 14

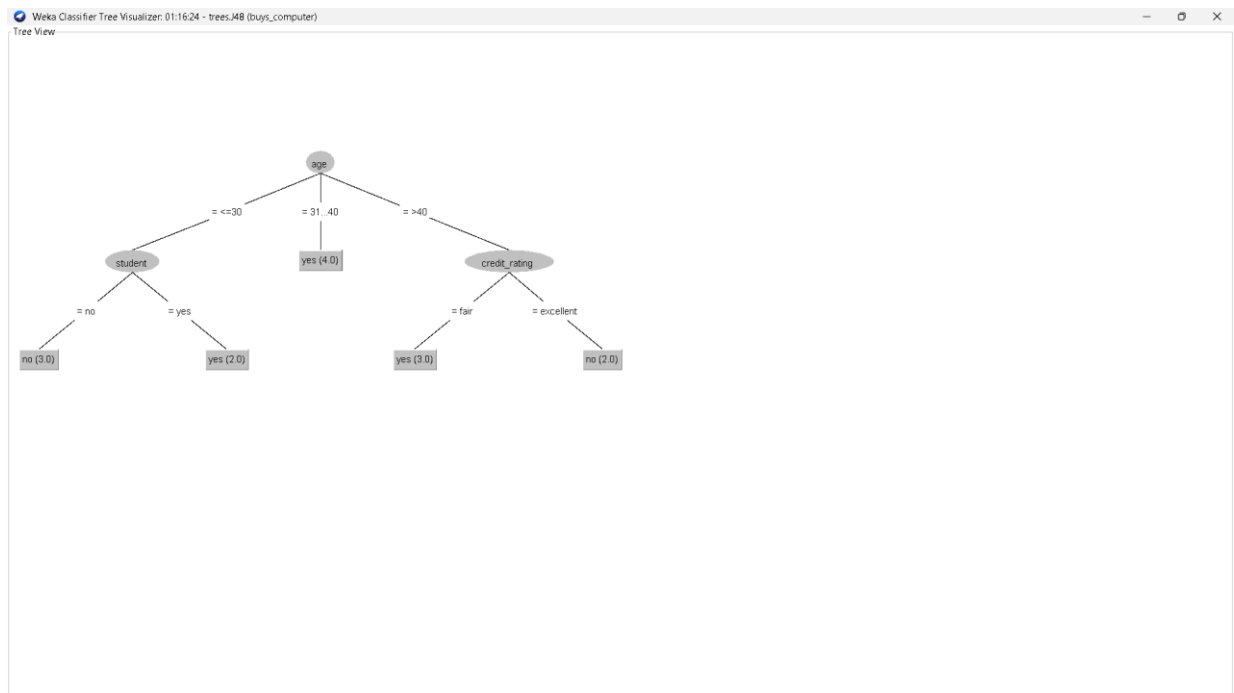
=== Detailed Accuracy By Class ===
          TP Rate  FP Rate  Precision  Recall  F-Measure  MCC  ROC Area  PRC Area  Class
0.400  0.444  0.333  0.400  0.364  -0.043  0.633  0.457  no
0.556  0.600  0.625  0.556  0.588  -0.043  0.633  0.758  yes
Weighted Avg.  0.500  0.544  0.521  0.500  0.508  -0.043  0.633  0.650

=== Confusion Matrix ===
 a b <-- classified as
 2 3 | a = no
 4 5 | b = yes

```

Status: OK

Log



Question 4: Linear Regression and Multiple Regression

Software / Environment: R Studio (or WEKA LinearRegression)

Aim:

To analyze the diabetes trend for different age groups by fitting both a Simple Linear Regression model and a Multiple Linear Regression model on the diabetes.csv dataset.

Algorithm:

1. Load the diabetes dataset (or initialize a sample dataframe containing Age, BloodPressure, and Glucose).
2. Fit a Simple Linear Regression model using `lm(Glucose ~ Age)` to model glucose levels directly based on age.
3. Fit a Multiple Linear Regression model using `lm(Glucose ~ Age + BloodPressure)` to examine how glucose is affected jointly by age and blood pressure.
4. Display regression coefficients, standard errors, and R^2 values using the `summary()` function.

Program (R Code)

```
R
# Sample data corresponding to diabetes.csv
diabetes <- data.frame(
  Age = c(21, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70),
  BloodPressure = c(60, 65, 70, 72, 78, 82, 85, 88, 90, 92, 95),
  Glucose = c(85, 89, 95, 105, 115, 120, 135, 140, 155, 160, 170)
)
```

1. Simple Linear Regression

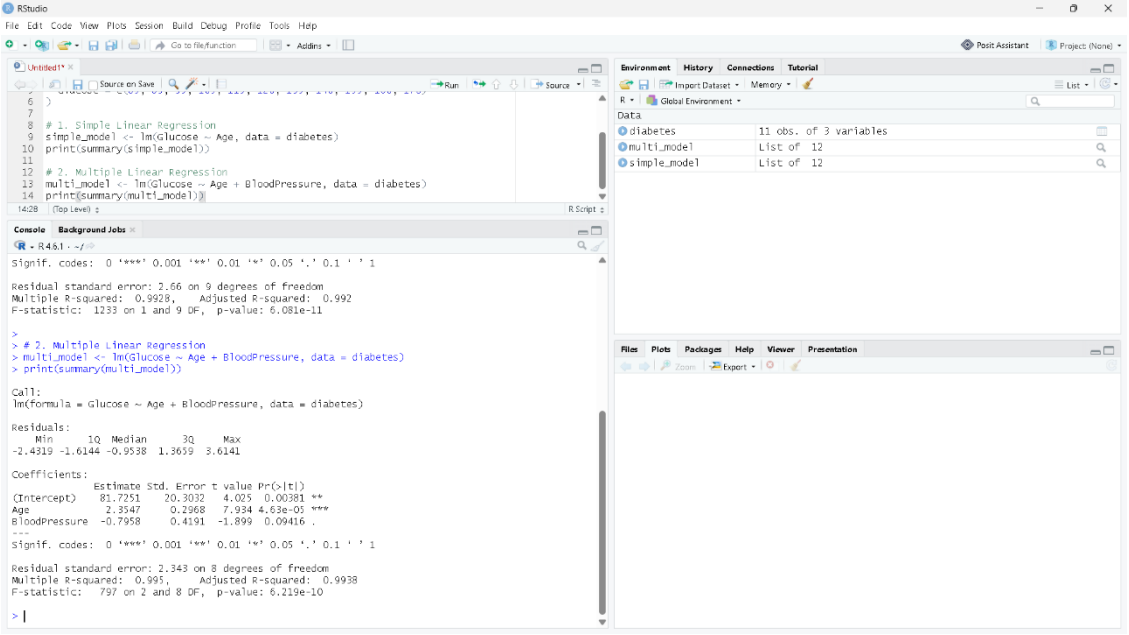
```
simple_model <- lm(Glucose ~ Age, data = diabetes)
```

```
print(summary(simple_model))
```

2. Multiple Linear Regression

```
multi_model <- lm(Glucose ~ Age + BloodPressure, data = diabetes)
```

```
print(summary(multi_model))
```



The screenshot shows the RStudio interface with the following components:

- Script Editor:** Contains the R code for simple and multiple linear regression.
- Environment:** Lists the objects in the global environment: `diabetes` (11 obs. of 3 variables), `multi_model` (List of 12), and `simple_model` (List of 12).
- Console:** Displays the output of the R commands, including the summary of the simple linear regression model.

```
6  
7  
8 # 1. Simple Linear Regression  
9 simple_model <- lm(Glucose ~ Age, data = diabetes)  
10 print(summary(simple_model))  
11  
12 # 2. Multiple Linear Regression  
13 multi_model <- lm(Glucose ~ Age + BloodPressure, data = diabetes)  
14 print(summary(multi_model))  
1428 (Top Level) >
```

Console Output:

```
R451 ~ /  
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
Residual standard error: 2.66 on 9 degrees of freedom  
Multiple R-squared: 0.9928, Adjusted R-squared: 0.992  
F-statistic: 1233 on 1 and 9 DF, p-value: 6.081e-11  
>  
> # 2. Multiple Linear Regression  
> multi_model <- lm(Glucose ~ Age + BloodPressure, data = diabetes)  
> print(summary(multi_model))  
Call:  
lm(formula = Glucose ~ Age + BloodPressure, data = diabetes)  
Residuals:  
    Min       1Q   Median       3Q      Max  
-2.4319 -1.6144 -0.9538  1.3659  3.6141  
Coefficients:  
            Estimate Std. Error t value Pr(>|t|)  
(Intercept)  81.7251    20.3032   4.025 0.00381 **  
Age           2.3547     0.2968   7.934 4.63e-05 ***  
BloodPressure -0.7958     0.4191  -1.899 0.09416 .  
---  
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
Residual standard error: 2.343 on 8 degrees of freedom  
Multiple R-squared: 0.995, Adjusted R-squared: 0.9938  
F-statistic: 797 on 2 and 8 DF, p-value: 6.219e-10  
> |
```

Question 5: WEKA Apriori & FP-Growth on T1 to T5

Software / Environment: WEKA Explorer (Preprocess & Associate tabs)

Aim:

To find all frequent itemsets using the Apriori algorithm and build the FP-Growth tree in WEKA for transactions T1 to T5 with $\text{min_sup} = 50\%$ (count ≥ 2 out of 5 transactions, or 0.4) and $\text{min_conf} = 80\%$ (0.8).

Transactions Given:

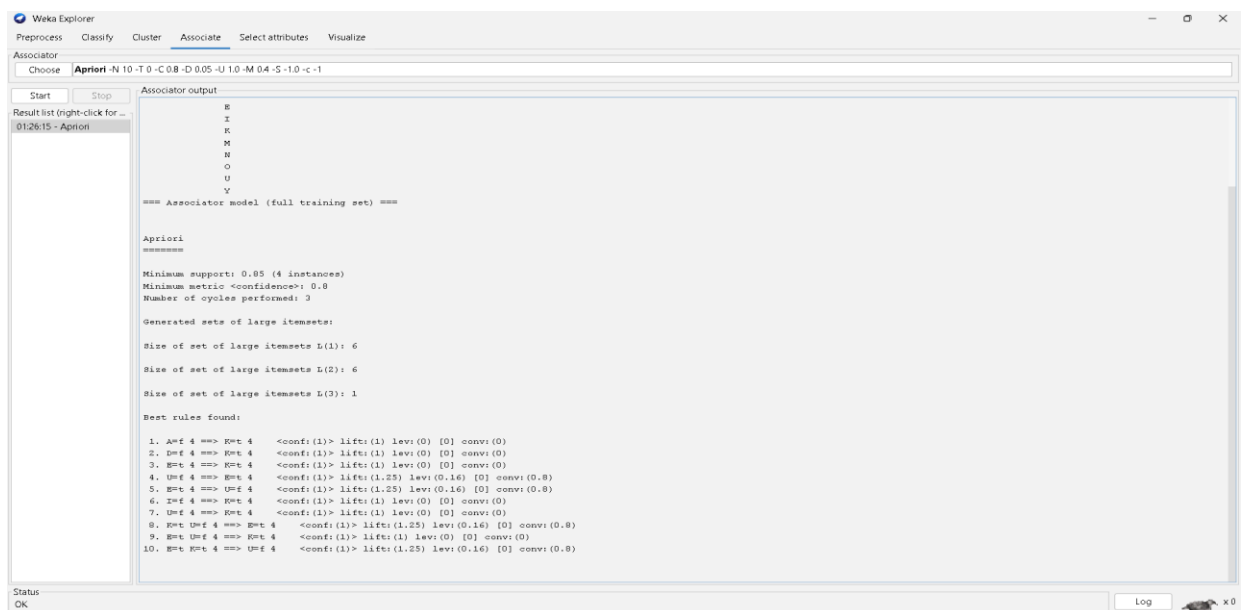
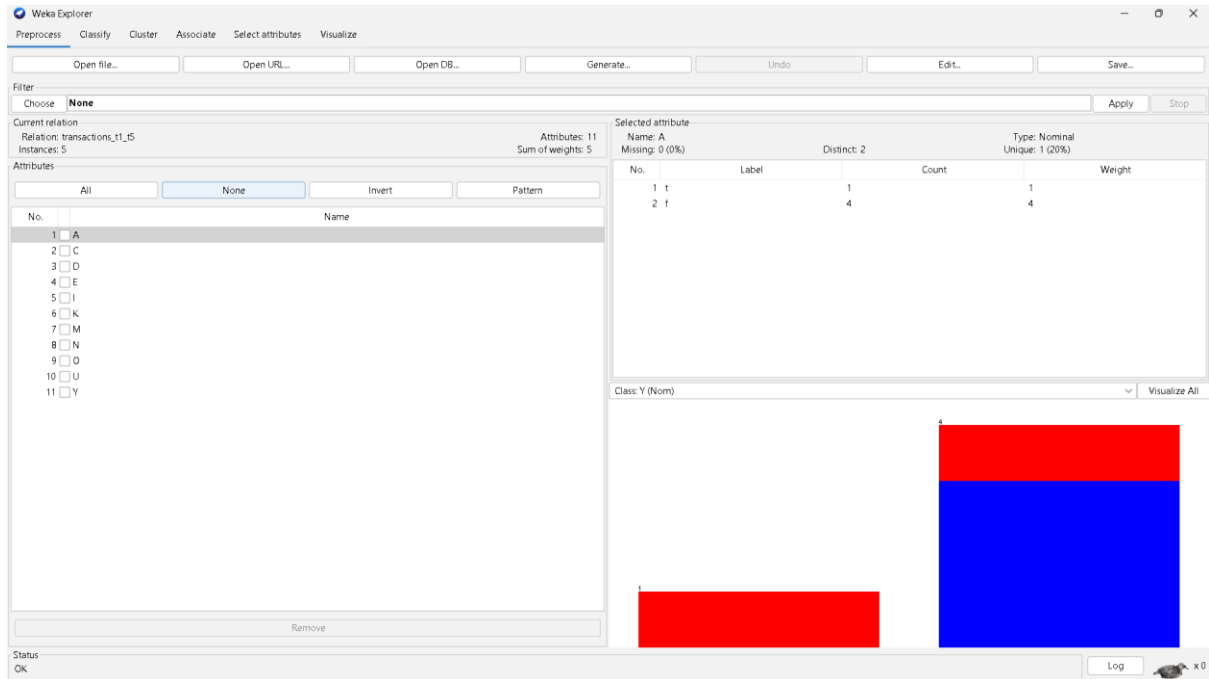
- T1: {M, O, N, K, E, Y}
- T2: {D, O, N, K, E, Y}
- T3: {M, A, K, E}
- T4: {M, U, C, K, Y}
- T5: {C, O, O, K, I, E}

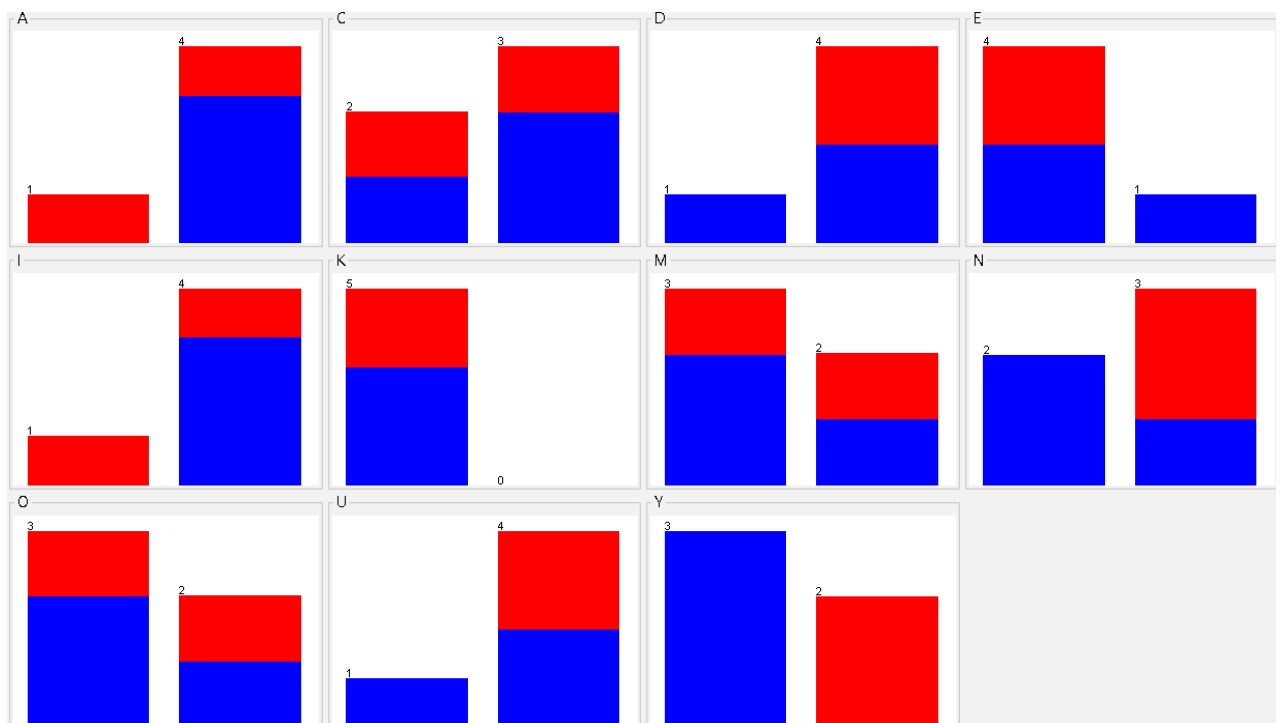
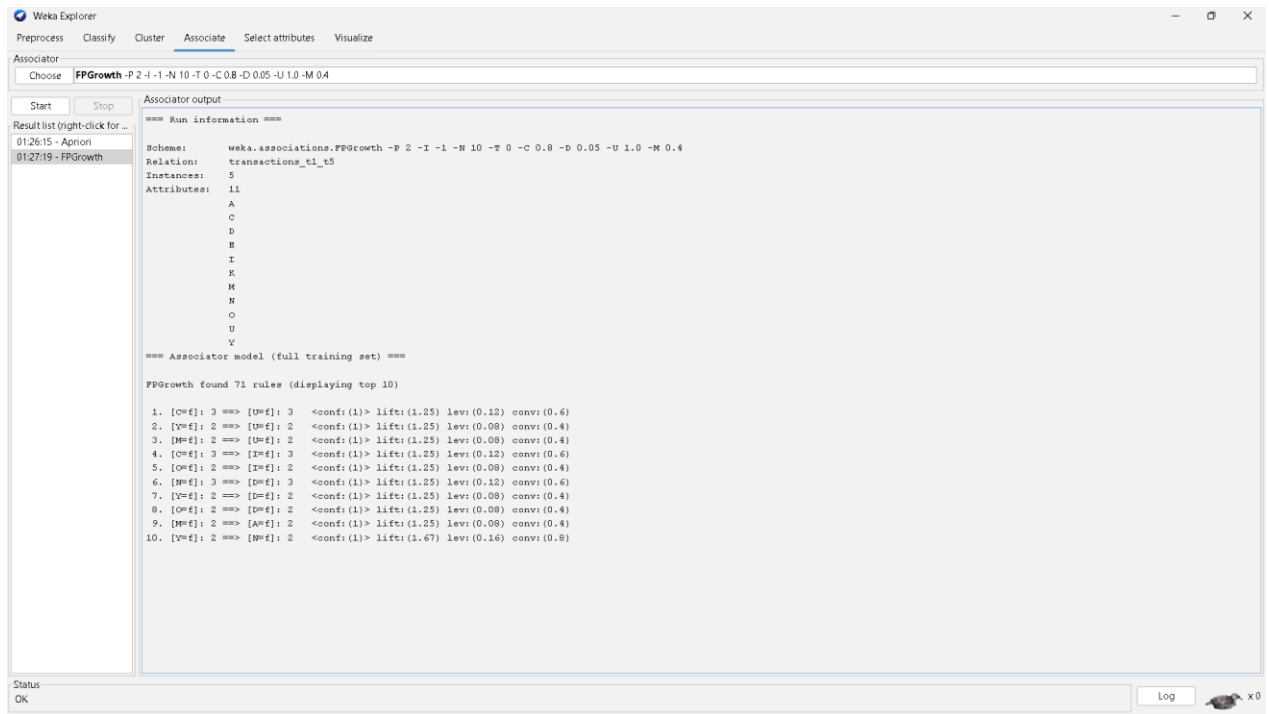
Procedure (Steps in WEKA):

1. Create an ARFF file named transactions_t1_t5.arff with binary nominal attributes ($\{t, f\}$) for all unique items: A, C, D, E, I, K, M, N, O, U, Y.
2. Open **WEKA Explorer**, click **Open file...** under the **Preprocess** tab, and select transactions_t1_t5.arff.
3. Go to the **Associate** tab.
4. **Apriori:**
 - Click the text box next to **Choose** (where Apriori is displayed).
 - Set lowerBoundMinSupport = 0.4 (corresponding to ≥ 2 instances).
 - Set minMetric (confidence) = 0.8.
 - Click **OK**, then click **Start**.

5. FP-Growth:

- Click **Choose** → select FPGrowth.
- Set lowerBoundMinSupport = 0.4 and minMetric = 0.8.
- Click **Start**.





Question 6: Prediction of Categorical Data using Decision Tree and Logistic in WEKA

Software / Environment: WEKA Explorer (Preprocess & Classify tabs)

Aim:

To perform categorical data prediction by preprocessing the dataset, building a Decision Tree (J48), and evaluating with a Logistic Regression classifier (Logistic) in WEKA.

Procedure (Steps in WEKA):

1. Create or use a standard categorical dataset file (such as contact-lenses.arff from WEKA's data/ folder or buys_computer.arff).
2. **Preprocess:**
 - Open **WEKA Explorer** → Click **Open file...** under the **Preprocess** tab and select the .arff dataset.
 - Observe attribute distributions, distinct values, and class balances.
3. **Decision Tree (Tree):**
 - Navigate to the **Classify** tab.
 - Click **Choose** → classifiers → trees → select J48.
 - Select **Cross-validation (10 folds)** or **Use training set** under *Test options*.
 - Click **Start** to build the decision tree.
 - Right-click the result in the *Result list* and select **Visualize tree** to view the graphical tree.
4. **Logistic Regression (Logistic):**
 - On the **Classify** tab, click **Choose** → classifiers → functions → select Logistic.
 - Click **Start** and record the classification accuracy and confusion matrix.

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Choose **None** Apply Stop

Current relation: Relation: weather.nominal Instances: 14 Attributes: 5 Sum of weights: 14

Attributes: All None Invert Pattern

No.	Name
1	outlook
2	temperature
3	humidity
4	windy
5	play

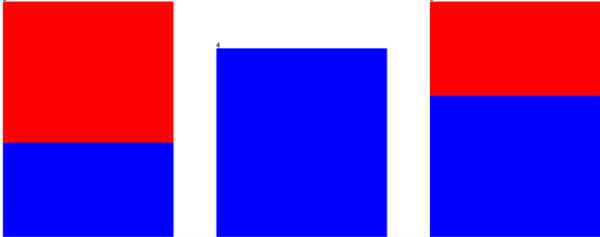
Remove

Status: OK

Selected attribute: Name: outlook Missing: 0 (0%) Distinct: 3 Type: Nominal Unique: 0 (0%)

No.	Label	Count	Weight
1	sunny	5	5
2	overcast	4	4
3	rainy	5	5

Class: play (Nom) Visualize All



Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **J48 -C 0.25 -M 2**

Test options: ☐ Use training set ☐ Supplied test set ☒ Cross-validation Folds: 10 ☐ Percentage split %: 66 More options...

(Nom) play Start Stop

Result list (right-click for options): 01:34:33 - trees.J48

Classifier output:

```
I humidity = high: no (3.0)
I humidity = normal: yes (2.0)
outlook = overcast: yes (4.0)
outlook = rainy
I windy = TRUE: no (2.0)
I windy = FALSE: yes (3.0)

Number of Leaves : 5
Size of the tree : 8

Time taken to build model: 0 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances      7      50 %
Incorrectly Classified Instances    7      50 %
Kappa statistic                    -0.0426
Mean absolute error                 0.4167
Root mean squared error             0.5984
Relative absolute error             87.5 %
Root relative squared error         121.2987 %
Total Number of Instances          14

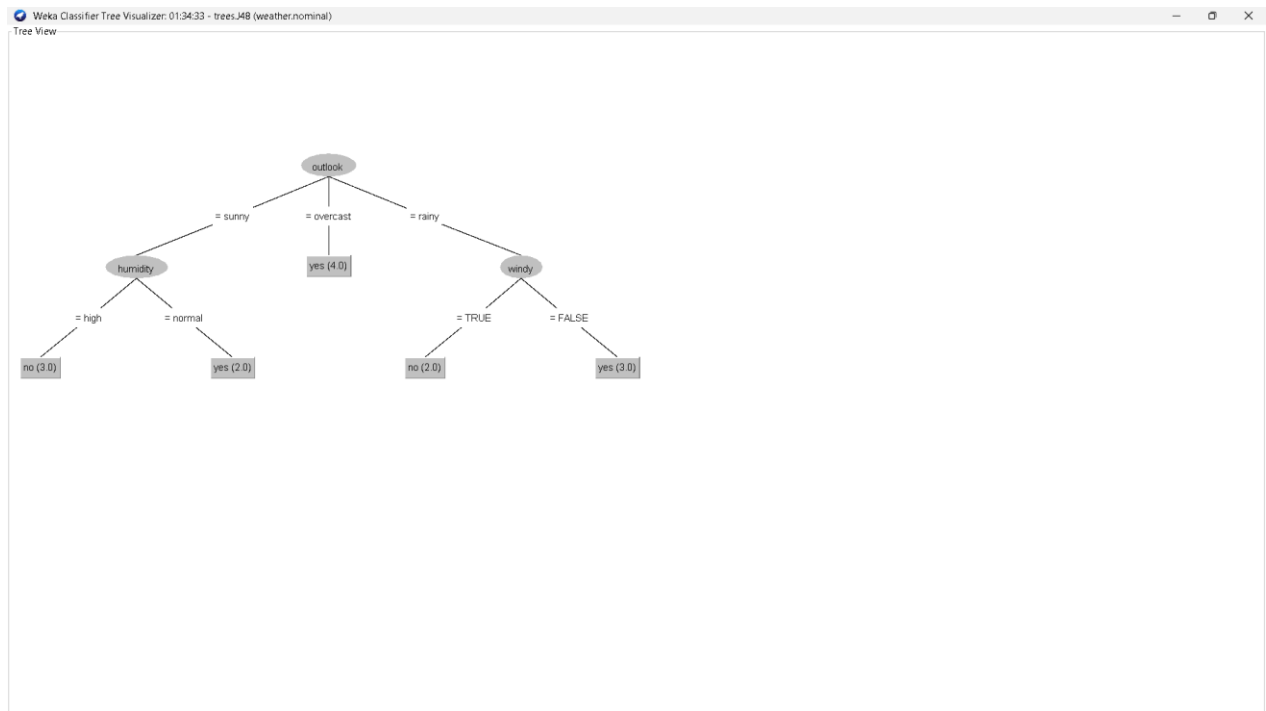
=== Detailed Accuracy By Class ===

      TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
0.556   0.600   0.625   0.556   0.588   -0.043   0.633   0.758   yes
0.400   0.444   0.333   0.400   0.364   -0.043   0.633   0.457   no
Weighted Avg.   0.500   0.544   0.521   0.500   0.508   -0.043   0.633   0.650

=== Confusion Matrix ===

a b  <-- classified as
5 4 | a = yes
3 2 | b = no
```

Status: OK



Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier: Choose **Logistic -R 1.0E-8 -M -1 -num-decimal-places 4**

Test options

- ☐ Use training set
- ☐ Supplied test set
- ☒ Cross-validation Folds: 10
- ☐ Percentage split %: 66

(Nom) play

Start Stop

Result list (right-click for options)

- 01:34:33 - trees.J48
- 01:35:54 - functionsLogistic

Classifier output

Variable yes

```
=====
outlook=sunny 0
outlook=overcast 9.73275593611619E24
outlook=rainy 0.0027
temperature=hot 0.0002
temperature= mild 3.0207875213740724E19
temperature=cool 0
humidity=normal 2.0353933107400553E51
windy=FALSE 4.877521304260806E31
=====
```

Time taken to build model: 0.01 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	10	71.4286 %
Incorrectly Classified Instances	4	28.5714 %
Kappa statistic	0.3778	
Mean absolute error	0.2858	
Root mean squared error	0.5345	
Relative absolute error	60.009 %	
Root relative squared error	108.3425 %	
Total Number of Instances	14	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
Weighted Avg.	0.778	0.400	0.600	0.600	0.600	0.378	0.767	0.655	yes
	0.600	0.222	0.600	0.600	0.600	0.378	0.767	0.655	no

=== Confusion Matrix ===

```
a b <-- classified as
7 2 | a = yes
2 3 | b = no
```

Status: OK

Log

Question 7: Association Rule Mining on Transaction ARFF Dataset

Software / Environment: WEKA Explorer (Associate tab) and Manual Verification

Aim:

To create an ARFF file for the given market basket transactions (T1 to T6), extract frequent itemsets, and generate accepted/rejected association rules using the Apriori and FP-Growth algorithms with minimum support threshold $s = 33.33\%$ (support count ≥ 2) and minimum confidence threshold $c = 60\%$.

Given Transactions Table:

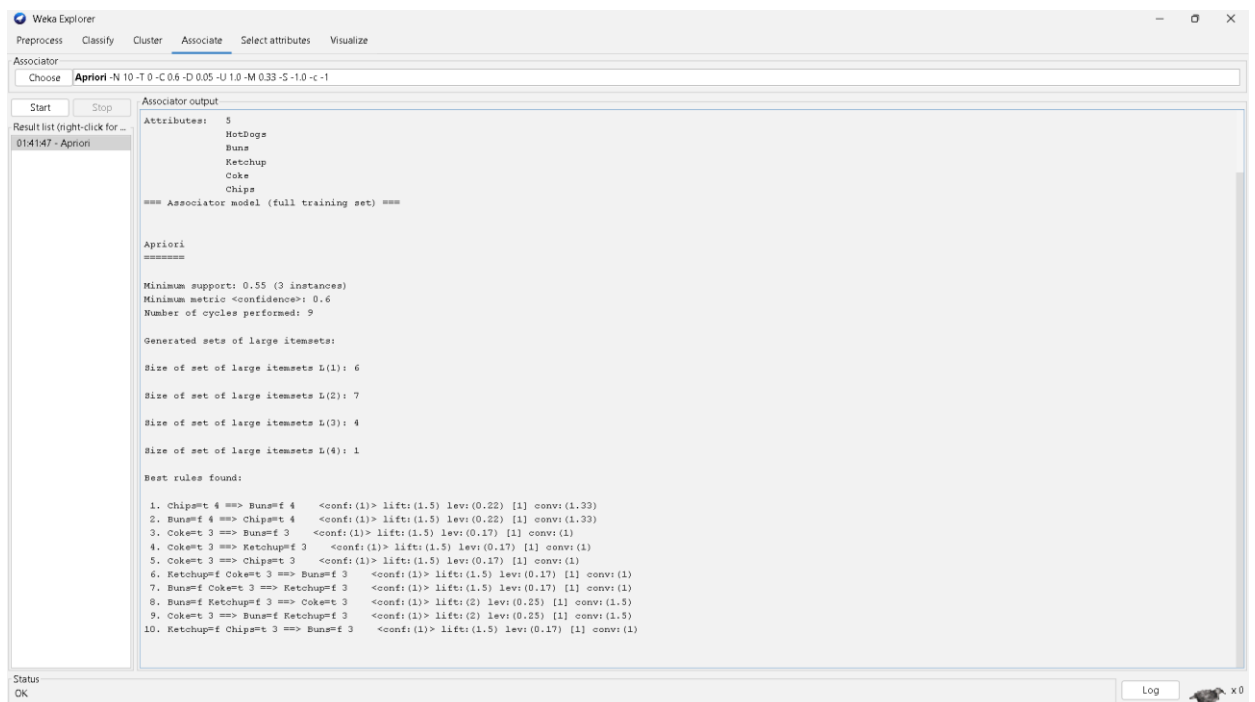
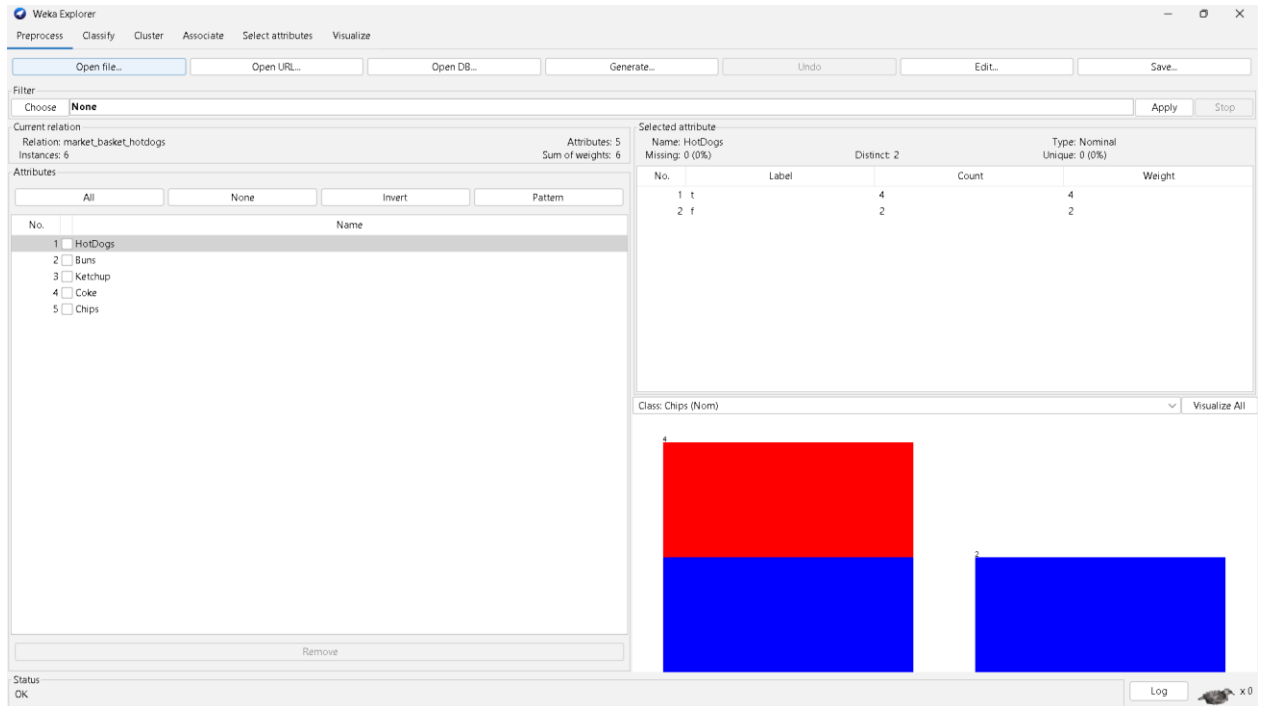
- T1: {Hot Dogs, Buns, Ketchup}
- T2: {Hot Dogs, Buns}
- T3: {Hot Dogs, Coke, Chips}
- T4: {Chips, Coke}
- T5: {Chips, Ketchup}
- T6: {Hot Dogs, Coke, Chips}

Procedure (Steps in WEKA):

1. Save the dataset below as hotdogs_transactions.arff.
2. Open **WEKA Explorer** → **Preprocess** → Click **Open file...** and select hotdogs_transactions.arff.
3. Go to the **Associate** tab.
4. Click the text box next to **Choose** to edit parameters:
 - Set lowerBoundMinSupport = 0.33 (corresponding to $\geq 2/6$ transactions).
 - Set minMetric (confidence) = 0.60 (60% threshold).

- Set numRules = 10.
- Click **OK**.

5. Click **Start** to run the algorithm and view the extracted association rules.



Question 8: Rule-Based vs Decision Tree Classification & Comparison

Software / Environment: WEKA Explorer (Classify tab) and R Studio (for plotting the comparison graph)

Aim:

To perform categorical data classification using Rule-based classification (JRip / PART) and Decision Tree classification (J48) in WEKA, compare their classification accuracy, and plot a comparative bar chart.

Procedure (Steps in WEKA):

1. Open **WEKA Explorer** and click **Open file...** under the **Preprocess** tab to load a categorical dataset (e.g., weather.nominal.arff or breast-cancer.arff located in the WEKA data/ folder).
2. Go to the **Classify** tab.
3. **Decision Tree Classification (J48):**
 - Click **Choose** → classifiers → trees → select J48.
 - Select **Cross-validation (10 folds)** under *Test options*.
 - Click **Start** and note down the percentage of **Correctly Classified Instances** (Accuracy).
4. **Rule-Based Classification (JRip):**
 - Click **Choose** → classifiers → rules → select JRip (or PART).
 - Select **Cross-validation (10 folds)** under *Test options*.
 - Click **Start** and note down the percentage of **Correctly Classified Instances** (Accuracy).
5. Compare the accuracy results of both models and plot the comparative bar chart in R Studio.

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Choose **None** Apply Stop

Current relation: Relation: weather.nominal Instances: 14 Attributes: 5 Sum of weights: 14

Attributes: All None Invert Pattern

No. Name

1 outlook
2 temperature
3 humidity
4 windy
5 play

Remove

Status: OK Log x0

Selected attribute: Name: outlook Missing: 0 (0%) Distinct: 3 Type: Nominal Unique: 0 (0%)

No.	Label	Count	Weight
1	sunny	5	5
2	overcast	4	4
3	rainy	5	5

Class: play (Nom) Visualize All

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier: Choose **J48 -C 0.25 -M 2**

Test options: ☐ Use training set ☐ Supplied test set ☒ Cross-validation Folds: 10 ☐ Percentage split %: 66 More options...

(Nom) play Start Stop

Result list (right-click for options): 01:55:26 - trees.J48

Classifier output:

```
| humidity = high: no (3.0)
| humidity = normal: yes (2.0)
outlook = overcast: yes (4.0)
outlook = rainy
| windy = TRUE: no (2.0)
| windy = FALSE: yes (3.0)

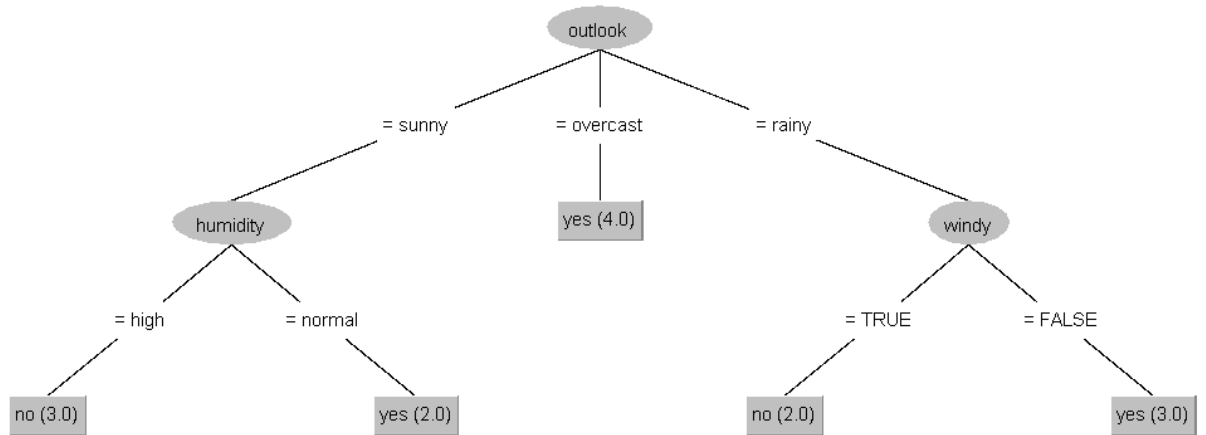
Number of leaves : 5
Size of the tree : 8
Time taken to build model: 0 seconds

=== Stratified cross-validation ===
=== Summary ===
Correctly Classified Instances      7      50 %
Incorrectly Classified Instances    7      50 %
Kappa statistic                    -0.0426
Mean absolute error                 0.4167
Root mean squared error             0.5984
Relative absolute error             87.5 %
Root relative squared error        121.2987 %
Total Number of Instances          14

=== Detailed Accuracy By Class ===
               TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
0.556    0.600    0.625    0.556    0.588    -0.043    0.633    0.758    yes
0.400    0.444    0.333    0.400    0.364    -0.043    0.633    0.457    no
Weighted Avg.  0.500    0.544    0.521    0.500    0.508    -0.043    0.633    0.650

=== Confusion Matrix ===
a b  <-- classified as
5 4 | a = yes
3 2 | b = no
```

Status: OK Log x0



Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **JRip - F 3 - N 2.0 - O 2 - S 1**

Test options:
☐ Use training set
☐ Supplied test set
☒ Cross-validation Folds: **10**
☐ Percentage split %: **66**
 More options...

(Nom) play
 Start Stop

Result list (right-click for options):
 01:55:26 - trees.J48
01:56:43 - rules.JRip

Classifier output

=== Classifier model (full training set) ===

JRIP rules:
 =====
 (humidity = high) and (outlook = sunny) => play=no (3.0/0.0)
 (outlook = rainy) and (windy = TRUE) => play=no (2.0/0.0)
 => play=yes (9.0/0.0)
 =====

Number of Rules : 3

Time taken to build model: 0 seconds

=== Stratified cross-validation ===
 === Summary ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
Correctly Classified Instances	9					64.2857 %			
Incorrectly Classified Instances	5					35.7143 %			
Kappa statistic						0.186			
Mean absolute error						0.3674			
Root mean squared error						0.6038			
Relative absolute error						77.1635 %			
Root relative squared error						108.1927 %			
Total Number of Instances						14			

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
Weighted Avg.	0.778	0.400	0.700	0.778	0.737	0.189	0.656	0.757	yes
	0.400	0.222	0.500	0.400	0.444	0.189	0.656	0.646	no
	0.643	0.465	0.629	0.643	0.632	0.189	0.656	0.717	

=== Confusion Matrix ===

```

a b   <-- classified as
7 2 | a = yes
3 2 | b = no
  
```

Status: OK

Log